

Questions with Answer Keys

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Q1: 24 Feb (Shift 1) - Single Correct

Consider the elements Mg, Al, S, P and Si, the correct increasing order of their first ionization enthalpy is:

- (1) $\text{Al} < \text{Mg} < \text{Si} < \text{S} < \text{P}$
- (2) $\text{Al} < \text{Mg} < \text{S} < \text{Si} < \text{P}$
- (3) $\text{Mg} < \text{Al} < \text{Si} < \text{S} < \text{P}$
- (4) $\text{Mg} < \text{Al} < \text{Si} < \text{P} < \text{S}$

Q2: 25 Feb (Shift 2) - Single Correct

The correct order of bond dissociation enthalpy of halogen is :

- (1) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$
- (2) $\text{Cl}_2 > \text{F}_2 > \text{Br}_2 > \text{I}_2$
- (3) $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$
- (4) $\text{I}_2 > \text{Br}_2 > \text{Cl}_2 > \text{F}_2$

Q3: 26 Feb (Shift 1) - Single Correct

Match List-I with List-II

List -I

Electronic configuration of elements

List-II

$\Delta_i H$ in kJ mol^{-1}

- | | |
|----------------------|------------|
| (a) $1s^2 2s^2$ | (i) 801 |
| (b) $1s^2 2s^2 2p^4$ | (ii) 899 |
| (c) $1s^2 2s^2 2p^3$ | (iii) 1314 |
| (d) $1s^2 2s^2 2p^1$ | (iv) 1402 |

- (1) (a) – (ii), (b) – (iii), (c) – (iv), (d) – (i)
- (2) (a) – (iv), (b) – (i), (c) – (ii), (d) – (iii)
- (3) (a) – (i), (b) – (iv), (c) – (iii), (d) – (ii)
- (4) (a) – (i), (b) – (iii), (c) – (iv), (d) – (ii)

Q4: 26 Feb (Shift 2) - Single Correct

Given below are two statements; one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: In TlI_3 , isomorphous to CsI_3 , the metal is present in +1 oxidation state.

Reason R: Tl metals has fourteen f electrons in its electronic configuration.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both A and R are correct and R is the correct explanation of A
- (2) A is not correct but R is correct
- (3) Both A and R are correct R is NOT the correct explanation of A
- (4) A is correct but R is not correct

Q5: 26 Feb (Shift 2) - Single Correct

The correct order of electron gain enthalpy is:

- (1) $\text{S} > \text{Se} > \text{Te} > \text{O}$
- (2) $\text{O} > \text{S} > \text{Se} > \text{Te}$
- (3) $\text{S} > \text{O} > \text{Se} > \text{Te}$
- (4) $\text{Te} > \text{Se} > \text{S} > \text{O}$

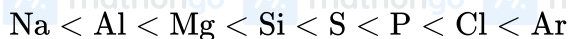
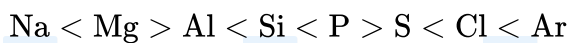
Answer Key**Q1 (1)****Q2 (3)****Q3 (1)****Q4 (3)****Q5 (1)**

Hints and Solutions

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Q1 (1)

Order of IE, in 3rd period is

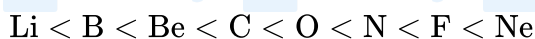


Q2 (3)

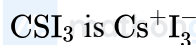
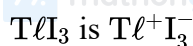
Fact based F_2 has $\text{F} - \text{F}$, F_2 involves repulsion of non-bonding electrons & more over its size is small & hence due to high repulsion its bond dissociation energy is very low.

Q3 (1)

Order of I.E. in second period



Q4 (3)



Thallium shows Tl^+ state due to inert pair effect.

Q5 (1)

Electron gain enthalpy of O is very low due to small size.